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Peptide Stapling by Crosslinking Two Amines with α-Ketoaldehydes through Diverse Modified Glyoxal-Lysine Dimer Linkers

*Pan Guo, Xin Chu, Chengjin Wu, Tianjiao Qiao, Wenli Guan, Chuanzheng Zhou, Tao Wang, Changlin Tian, Gang He, and Gong Chen**

Abstract: α-Ketoaldehydes play versatile roles in the ubiquitous natural processes of protein glycation. However, leveraging the reactivity of α-ketoaldehydes for biomedical applications has been challenging. Previously, the reactivity of α-ketoaldehydes with guanidine has been harnessed to design probes for labeling Arg residues on proteins in an aqueous medium. Herein, a highly effective, broadly applicable, and operationally simple protocol for stapling native peptides by crosslinking two amino groups through diverse imidazolium linkers with various α-ketoaldehyde reagents is described. The use of hexafluoroisopropanol as a solvent facilitates rapid and clean reactions under mild conditions and enables unique selectivity for Lys over Arg. The naturally occurring GOLD/MOLD linkers have been expanded to encompass a wide range of modified glyoxal-lysine dimer (OLD) linkers. In a proof-of-concept trial, these modular stapling reactions enabled a convenient two-round strategy to streamline the structure–activity relationship (SAR) study of the wasp venom peptide anoplin, leading to enhanced biological activities.

Introduction

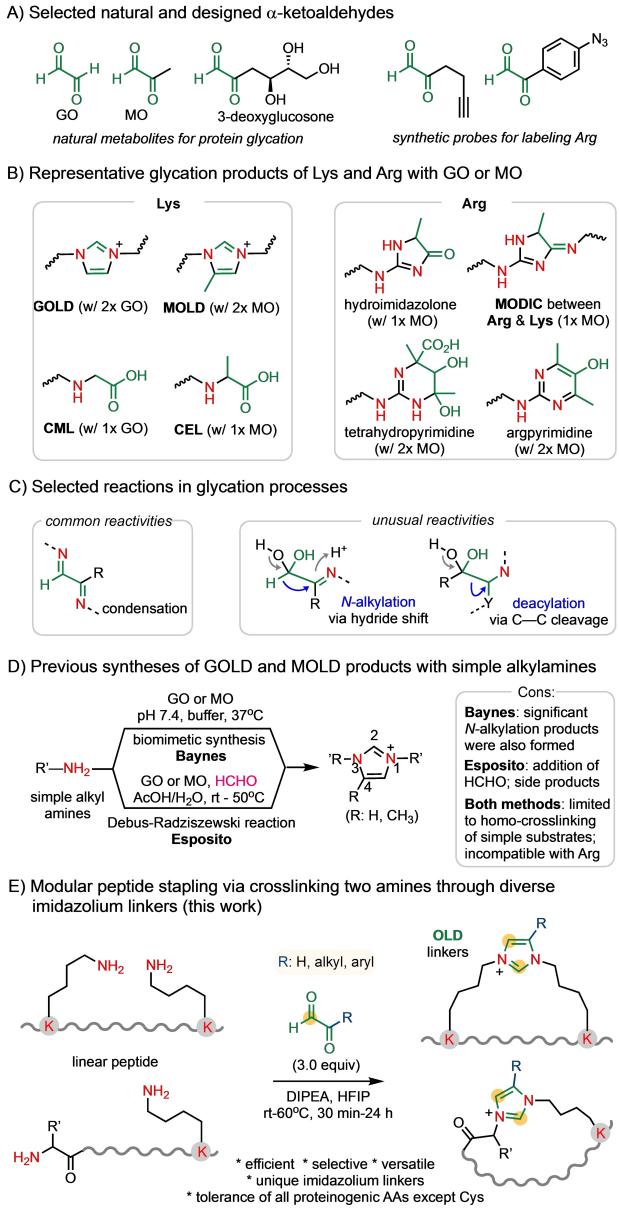
Peptide macrocycles have become a powerful platform for exploring new chemical spaces in drug discovery.[1] While significant progress has been made in chemistry for cyclizing pre-engineered peptide substrates equipped with exogenous reactive groups, methods capable of stapling native peptides through their endogenous side chains are much underexplored.[2] Presently, the strategies for stapling native peptides predominantly revolve around the reactivity of cysteine (Cys).[3,4] To broaden the toolkit, peptide modification techniques that selectively target non-Cys α amino acid (AA) residues are pressingly needed.[5–8]

As in many other areas of chemical research, nature could provide invaluable inspiration for developing tools to modify native peptides and proteins. Notably, nature employs various aldehyde metabolites to modify *N*-containing groups such as lysine (Lys) and arginine (Arg) side chains within biomolecules. One of the most prevalent aldehyde-mediated modification processes is protein glycation, where various α-ketoaldehydes derived from sugars, such as glyoxal (GO) and methylglyoxal (MO), react with proteins to form a wide spectrum of the so-called advanced glycation end products (Scheme 1A, 1B).[9–12] The vicinal dicarbonyl group gives α-ketoaldehydes unique reactivity, and allows them to react with various *N*-containing groups under physiological conditions without enzyme catalysis. The carbonyl condensation adducts of α-ketoaldehydes and N groups can undergo unusual bond rearrangements such as 1,2-hydride shift and C–C cleavage to generate a variety of products with interesting structures (Scheme 1C). For example, two molecules of GO or MO can react with two Lys residues to form the so-called imidazolium-linked glyoxal-lysine dimer (GOLD) and methylglyoxal-lysine dimer (MOLD) respectively.[10] The reactions of GO and MO with the amino side chain of Lys can also form *N*-alkylation products carboxymethyl or carboxyethyl lysine (CML, CEL).[11] Moreover, GO and MO can react with the guanidine side chain of Arg to form hydroimidazolone and argpyrimidine.[12a] They even can crosslink Lys and Arg with imidazolone linkers such as MODIC (Scheme 1B).[12b]

Despite the versatile reactivity displayed by α-ketoaldehydes in natural protein glycation processes, harnessing such reactivity for biomedical applications has been challenging. To date, success has been largely limited to the use of 1,2-dicarbonyl compounds for labeling Arg residues on proteins in an aqueous medium (Scheme 1A).[13] In comparison, the reactivity of α-ketoaldehydes with amines has only sporadically been explored.[10] In groundbreaking biomimetic investigations, Baynes demonstrated that GO or MO can react with simple alkylamines to form the corresponding GOLD/MOLD products in moderate yields under physiological conditions (Scheme 1D).[10a,b] However, the formation of *N*-alkylation side products was also significant. Later, Esposito introduced a more efficient protocol to prepare the GOLD/

[*] P. Guo, X. Chu, C. Wu, Dr. T. Qiao, W. Guan, Prof. C. Zhou, Prof. G. He, Prof. Dr. G. Chen
State Key Laboratory and Institute of Elemento-Organic Chemistry, College of Chemistry, Nankai University, Tianjin 300071, China
E-mail: gongchen@nankai.edu.cn
Prof. Dr. G. Chen
Frontiers Science Center for New Organic Matter, Nankai University, Tianjin 300071, China
Prof. Dr. G. Chen
Haihe Laboratory of Sustainable Chemical Transformations, Tianjin 300192, China
Dr. T. Wang, Prof. Dr. C. Tian
Hefei National Laboratory for Physical Sciences at the Microscale, University of Science and Technology of China, Hefei, 230026, China

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Scheme 1. Peptide stapling by crosslinking two amines with α-ketoaldehydes.

MOLD linkers in the mixed solvents of AcOH and H_2O using a Debus-Radziszewski reaction, where an additional formaldehyde (HCHO) reagent was required.[10c] Notably, both protocols were solely demonstrated on simple alkylamine substrates and the potential interference of Arg residue under their reaction conditions remains unaddressed.[13] Herein, we report the development of a highly effective, broadly applicable, and operationally simple protocol for stapling native peptides via crosslinking two amino groups with various α-ketoaldehydes through diverse imidazolium linkers (Scheme 1E). The use of hexafluoroisopropanol (HFIP) solvent facilitates rapid and clean reactions under mild conditions and enables unique selectivity for Lys over Arg. The naturally occurring GOLD/MOLD linkers are expanded to include a wide range of modified glyoxal-lysine dimer linkers, now referred to as OLD linkers. In a proof-of-concept trial, our modular stapling method provides a streamlined strategy for engineering the wasp venom peptide anoplin, enhancing its cytotoxicity against cancer cells.

Results and Discussion

Recently, we discovered that a simple HCHO reagent can crosslink a pair of Lys and tyrosine (Tyr) side chains on unprotected peptides through a methylene linker via a Mannich-type reaction in HFIP solvent at room temperature (rt).[8a] Notably, Arg side chain also shows high reactivity with HCHO and can crosslink with Lys through tetrahydrotriazine linkers.[8b] Encouraged by the unique promoting effect of HFIP solvent, we questioned whether other aldehyde reagents could also cross-link *N*-based AA side chains in high efficiency under mild conditions. Inspired by the MODIC linker in protein *N*-glycation,[12b] we initially tested the macrocyclization reaction of model tripeptide NA-Arg-Gly-Lys-NH_2 **1** (NA: 2-naphthoyl, for enhanced UV detection of peptides in LC–MS analysis) with glyoxal (GO) in HFIP at rt. Surprisingly, the reaction of **1** did not give any desired Arg-Lys crosslinked MODIC product, instead, forming the imidazolium-linked homo-dimerized GOLD compound **2** as the major product (see Supporting Information (SI) for details). Subsequently, we tested whether intramolecular crosslinking of two Lys with GO would give the GOLD-stapled product. To our delight, the reaction of model tetrapeptide NA-Lys-Gly-Ala-Lys-OH **3** at 10 mM concentration with 3.0 equiv. of GO and 3.0 equiv. of DIPEA base in HFIP solvent at rt for 30 min gave the desired product **4** in near quantitative HPLC yield (LC trace 2, Figure 1A). Notably, no *N*-alkylation side product **5** was formed; no condensation intermediates such as diimine **6** were observed by reverse phase LC–MS analysis. DIPEA was mainly used to neutralize the residual TFA in the peptide starting material. The use of other bases, such as K_2CO_3, can also work well. As in our HCHO-mediated crosslinking of Lys and Tyr, HFIP solvent was critical to achieving the high reactivity of the GO-mediated stapling. The same reaction in trifluoroethanol and MeOH gave the product **4** in 30% and 8% yield, respectively. Negligible conversion of **1** was observed in solvents like CH_3CN, DMF, or water (LC trace 3, see the Supporting Information for details). The imidazolium linker showed excellent stability in aqueous media of a wide range of pH (1–12).

As illustrated in Figure 1B, the GO-mediated HFIP-promoted stapling via two Lys side chains exhibited efficacy across a diverse range of peptide substrates with varying ring sizes and amino acid compositions under standard conditions [**A**]. The majority of stapled products were obtained with good to excellent LC yield and in reasonable HPLC isolated yield (shown in parathesis). However, in instances where the material loss occurred during HPLC separation, severely diminished isolated yields were observed for products with solubility issues (e.g. **17**, **19**). Except for Cys, all proteinogenic AA residues in unprotected form

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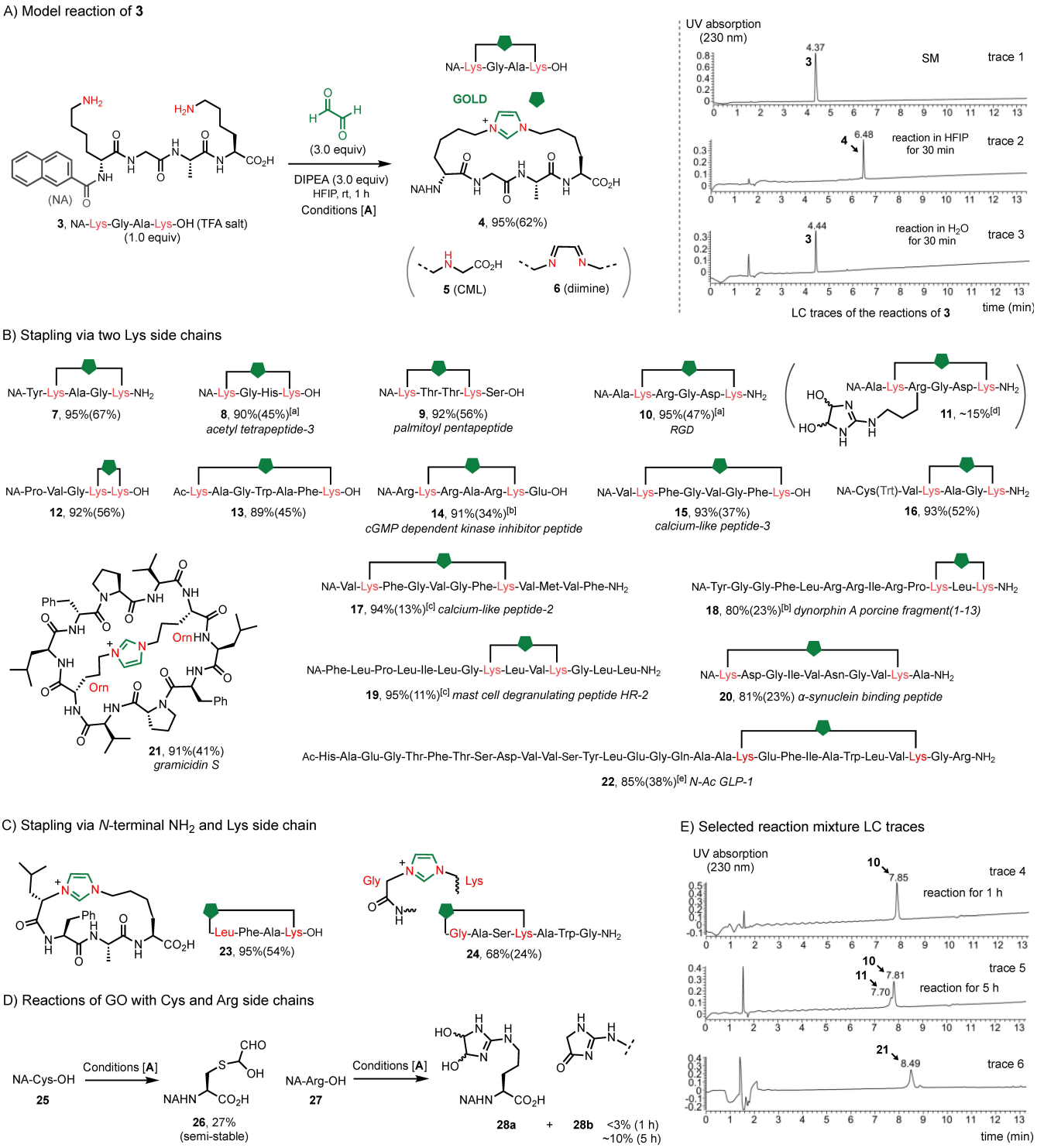


Figure 1. GO-mediated peptide stapling in HFIP. All reactions were conducted under the standard conditions at 10 mM concentration and a 0.02 mmol scale for 1 h unless otherwise specified. Yields are based on the UV absorption of peptide peaks in LC–MS traces. HPLC isolated yields are shown in parenthesis. [a] 5.0 equiv. of DIPEA. [b] 8.0 equiv. DIPEA. [c] 4.0 equiv. of GO. [d] 5 h. [e] 6.0 equiv. of DIPEA, 2 mM, 4 h.

proved tolerable in this reaction system. For example, Tyr (**7**), His (**8**), Trp (**13**), Thr (**9**), Ser (**9**), Asp (**10**), Asn (**20**), and Met (**17**) residues remained unaffected. The thiol side chain of NA-Cys-OH **25** rapidly reacted with GO, resulting in the formation of a semistable thiohemiacetal product **26** in a 27% yield, accompanied by smaller quantities of additional unidentified side products, as determined by LC–MS analysis (Figure 1D). The stapling of peptide substrates containing two Lys and an unprotected Cys led to a complex reaction mixture (see the Supporting Information for de-

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tails). In contrast, the stapling of peptides containing a Trt-protected Cys proceeded in high yield and selectivity (see **16**). While the guanidine side chain of Arg can react with GO to form a mixture of dihydroimidazole and hydroimidazolone under standard conditions, the reactivity of Arg is significantly lower than that of Lys, particularly in comparison to the stapling reaction involving two Lys. For instance, the reaction of NA-Arg-OH **27** with 3 equiv. of GO gave a mixture of **28a** and **28b**, with a less of 3 % yield in 1 h and around 10 % yield in 5 h (see the Supporting Information). As exemplified by compounds **10** and **14**, Arg-containing substrates can be stapled to give the GOLD products with excellent yield and chemoselectivity under standard conditions [**A**] with 3 equiv. of GO (see LC trace 4 in Figure 1E). The formation of Arg-derived side products can be mostly suppressed. However, the use of a larger excess of GO (e.g. 5 equiv) or an extended reaction time (e.g. 5 h) for Arg-containing substrates may result in the notable formation of dihydroimidazole side products (see the appearance of **11** in LC trace 5). In contrast to its high reactivity in our previously reported formaldehyde-mediated peptide modification in HFIP, Tyr exhibits no reactivity in the current GO-mediated reaction.[8a] The number of spacing AA units between the two Lys residues did not significantly influence the stapling efficiency (e.g. **12**, **13**). As shown by compounds **14, 15**, and **17**–**22**, a diverse range of bioactive linear peptides can react to give the corresponding stapled products in high yield. For example, the reaction of cyclic peptide antibiotic gramicidin S carrying two ornithine (orn) residues formed the bicyclic product **21** in 91 % LC yield (see LC trace 6). The reaction of 30-mer *N*-acetyl glucagon-like peptide-1 (GLP-1) formed the stapled product **22** in 85 % LC yield. As depicted by compounds **23** and **24**, the stapling between the *N*-terminal NH_2 groups and the Lys side chain also proceeded effectively under standard reaction conditions (Figure 1C).

As illustrated in Figures 2A and 2B, a diverse range of alkyl and aryl-substituted glyoxal derivatives can engage in analogous deacylative condensation reactions with two Lys side chains of linear peptides, forming the corresponding imidazolium-stapled products under similar conditions. Notably, C_4-substituted imidazolium linkers were generated as a regioisomeric mixture (see LC traces 7, 8). However, the regiochemistry of the imidazolium rings would not significantly alter the overall structure of the stapled peptides. α-Ketoaldehydes can be readily synthesized from easily accessible precursors.[14] A common method involves the SeO_2-mediated Riley oxidation of the corresponding methyl ketones, typically resulting in clean reactions with good yield (>70 %).[14] As evidenced by the reactions involving the model peptide **3**, methylglyoxal (MO) exhibited reactivity closely aligned with that of GO (see **4a**) (Figure 2B). The stapling reactivity might decrease with the size of the alkyl substituents on glyoxal (see **4b**–**4e**). Increasing the reaction temperature to 40 °C and using 4 equiv. of α-ketoaldehyde reagents usually restored the stapling efficiency. For example, the reaction of cyclohexyl glyoxal with peptide **3** at 40 °C for 4 h gave **4e** in near quantitative LC yield. In contrast, the reaction of adamantyl glyoxal **29** with **3** did not give any cyclized product. The steric hindrance imposed by the adamantyl group likely impeded the formation of the diimine intermediate or the subsequent deacylative condensation.

The reactions of aryl-substituted glyoxals necessitated slightly elevated conditions at 60 °C for 12 h. As illustrated by compounds **4f**–**4p**, various substituents on the phenyl ring were well tolerated and had a negligible impact on the stapling reactivity. The ethynyl tag on the phenyl ring remained unaffected (**4r**). Furyl (**4q**), naphthyl (**4s**), and pyrenyl (**4v**) groups can be incorporated into the imidazolium linker. Product **4t**, featuring a tetraphenylethene fluorophore, was formed in 84 % LC yield. Aryl glyoxal derived from the natural product celestolide also exhibited favorable reactivity (**4u**). It is noteworthy that aryl glyoxal reagents can undergo self-reaction, producing varying amounts of unidentified byproducts evident in LC traces. However, these side products do not interfere with the peptide stapling and can be easily removed through trituration treatment with diethyl ether or HPLC purification (see LC trace 9 for the reaction of **4j** and Supporting Information for details). Ethyl glyoxalate **30** did not give any cyclized product. The ethynyl group of compound **4r** could undergo a cycloaddition reaction with the azido group of Zidovudine **31**, resulting in product **32** in excellent yield under Cu-catalyzed conditions (Figure 2D). The stapling protocol with modified glyoxals also proved effective on complex peptide substrates (Figure 2E). For example, the reaction of 14-mer HR-2 peptide with MO and the reaction of palmitoyl peptide with 4-trifluoromethylphenyl glyoxal **34** yielded the corresponding products **19a** and **9h**, each in >90 % LC yields, respectively. The reaction of calcium-like peptide-2 with 4-ethynylphenyl glyoxal **33** gave **17r** in good yield.

As depicted in Figure 3A, the preparation of imidazolium-stapled peptides can be carried out on a solid phase with high efficiency. Linear peptide sequences bearing 1-(4,4-dimethyl-2,6-dioxocyclohex-1-ylidene)ethyl (Dde) protecting groups for specific Lys and acid-labile protecting groups for other AAs were first assembled on Rink amide resin using the standard solid phase peptide synthesis (SPPS) procedures. The Dde group on the resin was selectively removed by the treatment with 5 % $N_2H_4{\cdot}H_2O$ in DMF at rt; the resulting resin was then treated with excess amounts of α-ketoaldehyde and DIPEA in HFIP at varied temperatures. Finally, the treatment with TFA/H_2O removed all the acid-labile protecting groups and cleaved the stapled peptide from the resin. Notably, the Rink amide resin swelled reasonably well in HFIP and the imidazolium linkers were stable under TFA conditions. For example, GOLD-linked peptide **35** (**Ano-A**) was prepared in 34 % overall isolated yield with one HPLC purification.

As outlined in Figure 3B, our α-ketoaldehyde-mediated stapling reactions on solid phase enabled a convenient two-round strategy to streamline the SAR study of linear peptides for enhanced biological activities in a proof-of-concept investigation. Anoplin is a short amphipathic α-helical venom peptide with multiple pharmacological activities.[15] One recent study showed that anoplin can be

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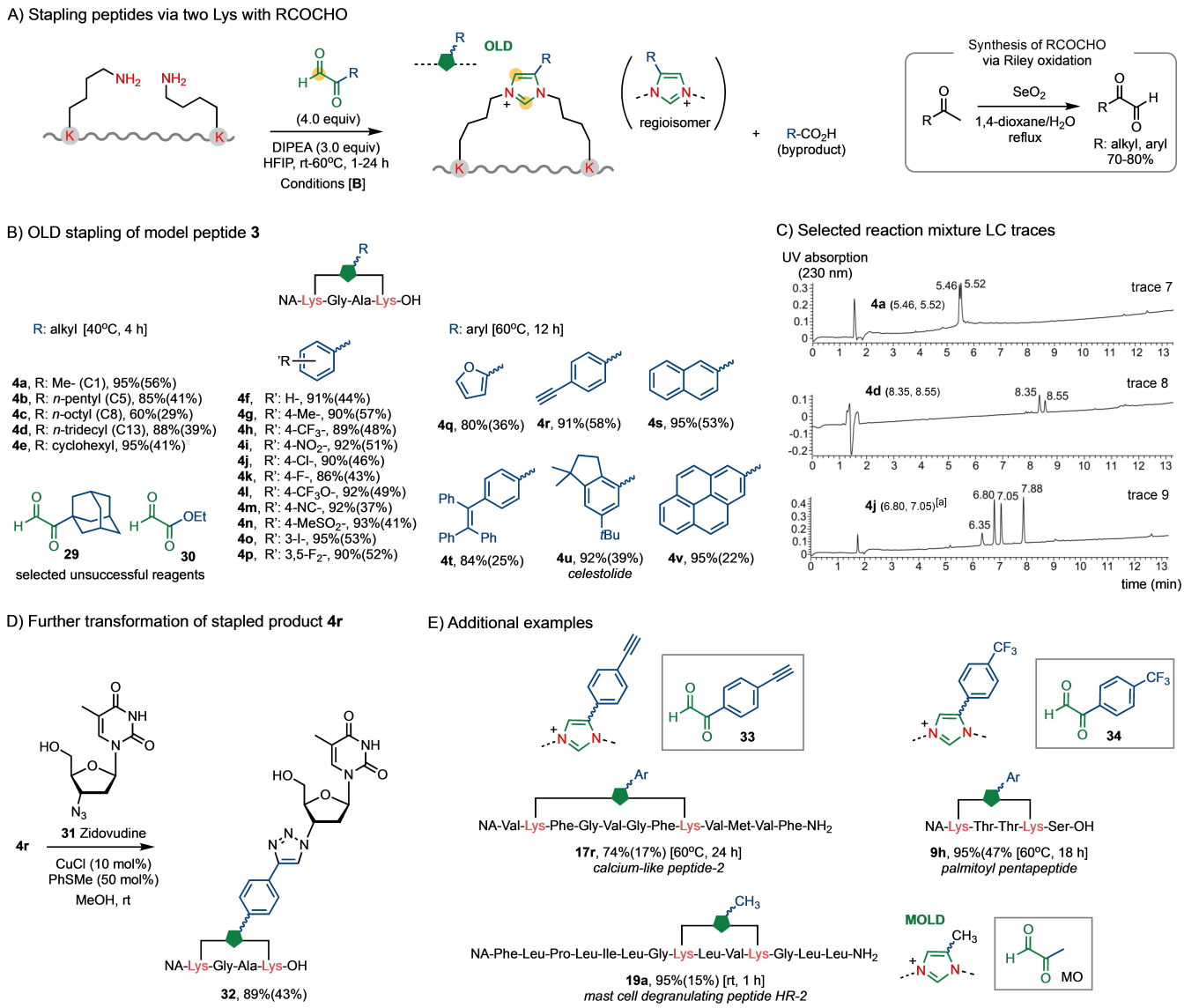


Figure 2. Peptide stapling with alkyl and aryl-substituted glyoxals. All reactions were conducted under the standard conditions at 0.5 mM concentration and 0.02 mmol scale for 1 h unless otherwise specified. Yields are based on the UV absorption of peptide peaks in LC–MS traces. HPLC isolated yields are shown in parenthesis. [a] Peaks at 6.35 and 7.88 min are non-peptide entities derived from the aryl glyoxal reagent. They can be readily removed by trituration with diethyl ether (see the Supporting Information for details).

stapled by metal-catalyzed alkene metathesis reactions to produce all hydrocarbon-stapled analogs with strong oncolytic activity against cancer cells.[16] However, the metathesis stapling method requires relatively sophisticated and costly synthetic manipulations. In our trial, anoplin was first mutated to **Ano** by replacing the non-essential C-terminal Leu with Phe for more sensitive UV detection in LC–MS analysis. In round one of our SAR study, two Lys residues were placed at various positions of **Ano**, and the resulting peptides were stapled with GO to give the corresponding GOLD-linked products. In round two, the GOLD-linker of selected analogs with enhanced activity was replaced with various OLD linkers using a series of modified glyoxal reagents (see Figure 2). Compounds **Ano-D** and **Ano-E** stood out in our first-round tests against mouse breast tumor 4T1 cells. Compounds **Ano-Dc** bearing an *n*-octyl substituent and **Ano-Eh** bearing a 4-trifluoromethylphenyl substituent were identified with further enhanced cytotoxicity in the second round of tests. Notably, both compounds showed close potency against 4T1 cells (IC_{50}: 6.6 and 6.8 μM respectively) to the most potent stapled peptides generated by the alkene metathesis method (IC_{50}: 4.0 μM).[16]

The mechanism of the GOLD/MOLD reactions has not been vigorously investigated in the literature. Based on the previous studies,[10a–c,11b] we propose a modified mechanistic model for the representative MOLD reaction under our newly optimized HFIP-promoted conditions (Scheme 2). The reaction presumably starts with the reversible condensation of MO with two alkylamines to give diimine **I**. Diimine **I** reacts with another molecule of MO to give

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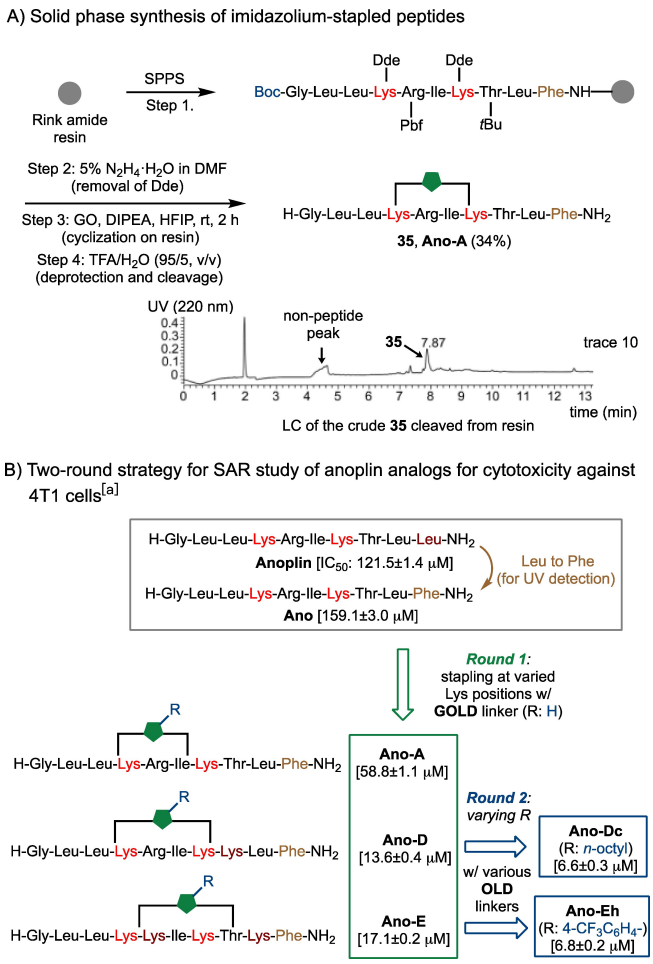
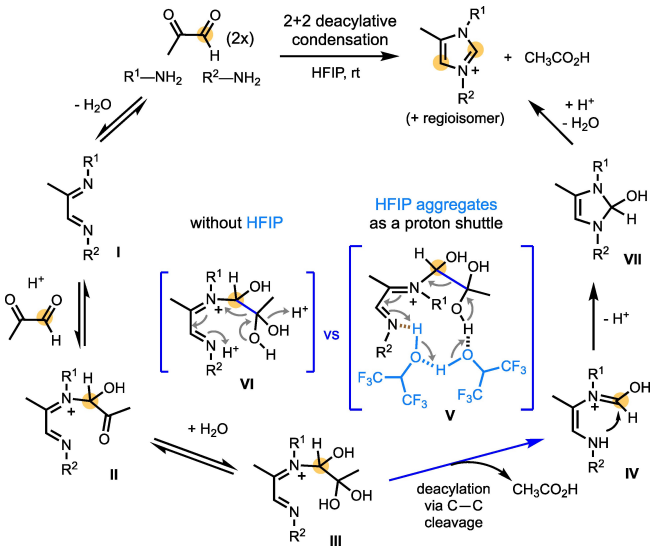


Figure 3. OLD stapling on solid phase enables a two-round strategy for SAR studies of anoplin for enhanced cytotoxicity. [a] The reported IC_{50} values are the mean ± standard deviation of three or more independent experiments. See the Supporting Information for the details of SPPS and cytotoxicity assays.



Scheme 2. Mechanistic considerations of HFIP-promoted MOLD reaction.

intermediate **II**, which then reacts with H_2O to give hydrate **III**. Deacylation of **III** via an unusual C–C cleavage gives intermediate **IV** and CH_3CO_2H. Intramolecular nucleophilic addition of **IV** followed by dehydration gives the final heteroaromatic imidazolium product, whose stability probably provides the driving force for the 2+2 deacylative condensation reaction. The C–C cleavage of **III** is likely the most difficult step of the entire reaction sequence. In general, the conformation of the 1,2-addition products of carbonyl compounds can strongly influence the elimination reactivity. The hyperconjugation interaction between the lone pair n orbital of the geminal OH group and the C–C σ* orbital might be critical for cleaving the C–C bond. The observed strong solvent dependence indicates that HFIP might play an intimate role in the critical deacylation step.[17] Notably, in addition to its acidic OH group, HFIP in solution is known to form relatively stable aggregates of up to 4 units, which possess even stronger H-bonding capabilities than single HFIP.[17c] In one of our recent studies, we discovered that HFIP aggregates can act as a dexterous proton shuttle to facilitate challenging transformations such as intermolecular hydride transfer via H-bond-mediated macrocyclic transition states.[8c] We suspect that HFIP aggregates might play a similar proton shuttle role in facilitating the deacylation step via H-bond-mediated macrocyclic transition states, in which optimal orbital interactions are allowed (see representative transition state **V** featuring an HFIP dimer proton shuttle).[18]

Conclusion

In summary, we developed a new method for stapling native peptides by crosslinking two amino groups through unique imidazolium linkers using various α-ketoaldehyde reagents. A variety of imidazolium crosslinkers can be installed with high efficiency and selectivity under mild conditions. The use of hexafluoroisopropanol (HFIP) solvent was key to achieving the high reactivity. As compared to the previously reported reactions in aqueous media, the reactions of α-ketoaldehydes in HFIP exhibit distinctive chemoselectivity for Lys over Arg. In a proof-of-concept trial, these modular stapling reactions enable a convenient two-round strategy to streamline the structure-and-activity study of venom peptide anoplin for oncolytic activity.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: peptide stapling · glycation · α-ketoaldehydes · solvent effects · imidazolium

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